



Equilibrium Index

Equilibrium index of an array is an index such that the sum of elements at lower indices is equal to the sum of elements at higher indices.

EXAMPLE

$$A = \{-7, 1, 5, 2, -4, 3, 0\}$$

3 is an equilibrium index, because:

$$A[0] + A[1] + A[2] = A[4] + A[5] + A[6]$$

6 is also an equilibrium index, because:

$$A[0] + A[1] + A[2] + A[3] + A[4] + A[5]$$

	0	1	2	3	4
A	1	3	5	2	2

pos = -1

leftSum = 0

rightSum = 0

Algorithm

```
for(i=0; i < N; i++)
    rightSum += A[i]

for(i=0; i < N; i++){
    rightSum -= A[i]
    if(leftSum == rightSum){
        pos = i + 1
        break
    }
    leftSum += A[i]
}
```

	0	1	2	3	4
A	1	3	5	2	2
	i				

pos = -1

leftSum = 0

rightSum = 1

Algorithm

```
for(i=0; i < N; i++)
```

```
    rightSum += A[i]
```

```
for(i=0; i < N; i++){
```

```
    rightSum -= A[i]
```

```
    if(leftSum == rightSum){
```

```
        pos = i + 1
```

```
        break
```

```
    }
```

```
    leftSum += A[i]
```

```
}
```

	0	1	2	3	4
A	1	3	5	2	2

pos = -1

leftSum = 0

rightSum = 13

Algorithm

```
for(i=0; i < N; i++)  
    rightSum += A[i]  
  
for(i=0; i < N; i++){  
    rightSum -= A[i]  
    if(leftSum == rightSum){  
        pos = i + 1  
        break  
    }  
    leftSum += A[i]  
}
```

	0	1	2	3	4
A	1	3	5	2	2
	i				

pos = -1

leftSum = 0

rightSum = 12

Algorithm

```
for(i=0; i < N; i++)
    rightSum += A[i]
```

```
for(i=0; i < N; i++){
    rightSum -= A[i]
    if(leftSum == rightSum){
        pos = i + 1
        break
    }
    leftSum += A[i]
}
```

	0	1	2	3	4
A	1	3	5	2	2
	i				

Is 0 == 12 ?

pos = -1

leftSum = 0

rightSum = 12

Algorithm

```
for(i=0; i < N; i++)
    rightSum += A[i]
```

```
for(i=0; i < N; i++){
    rightSum -= A[i]
    if(leftSum == rightSum){
        pos = i + 1
        break
    }
    leftSum += A[i]
}
```

	0	1	2	3	4
A	1	3	5	2	2
		i			

pos = -1

leftSum = 1

rightSum = 12

Algorithm

```
for(i=0; i < N; i++)
    rightSum += A[i]
```

```
for(i=0; i < N; i++){
    rightSum -= A[i]
    if(leftSum == rightSum){
        pos = i + 1
        break
    }
    leftSum += A[i]
}
```


	0	1	2	3	4
A	1	3	5	2	2
		i			

pos = -1

leftSum = 1

rightSum = 9

Algorithm

```
for(i=0; i < N; i++)  
    rightSum += A[i]
```

```
for(i=0; i < N; i++){  
    rightSum -= A[i]  
    if(leftSum == rightSum){  
        pos = i + 1  
        break  
    }  
    leftSum += A[i]  
}
```

	0	1	2	3	4
A	1	3	5	2	2
		<i>i</i>			

Is 1 == 9 ?

pos = -1

leftSum = 1

rightSum = 9

Algorithm

```
for(i=0; i < N; i++)
    rightSum += A[i]
```

```
for(i=0; i < N; i++){
    rightSum -= A[i]
    if(leftSum == rightSum){
        pos = i + 1
        break
    }
    leftSum += A[i]
}
```

	0	1	2	3	4
A	1	3	5	2	2

i

pos = -1

leftSum = 4

rightSum = 9

Algorithm

```
for(i=0; i < N; i++)  
    rightSum += A[i]
```

```
for(i=0; i < N; i++){  
    rightSum -= A[i]  
    if(leftSum == rightSum){  
        pos = i + 1  
        break  
    }  
    leftSum += A[i]  
}
```

	0	1	2	3	4
A	1	3	5	2	2
			i		

pos = -1

leftSum = 4

rightSum = 4

Algorithm

```
for(i=0; i < N; i++)
    rightSum += A[i]
```

```
for(i=0; i < N; i++){
    rightSum -= A[i]
    if(leftSum == rightSum){
        pos = i + 1
        break
    }
    leftSum += A[i]
}
```

```
1. //Max Equilibrium Sum
2. public class MES {
3.     public static int getMaxEquilibriumSumOptimized(int[] arr) {
4.         int totalSum = 0;
5.         int leftSum = 0;
6.         int maxSum = Integer.MIN_VALUE;
7.         for (int i = 0; i < arr.length; i++) {
8.             totalSum += arr[i];
9.         }
10.        for (int i = 0; i < arr.length; i++) {
11.
12.            totalSum -= arr[i];
13.            if (leftSum == totalSum && leftSum > maxSum) {
14.                maxSum = leftSum;
15.            }
16.            leftSum += arr[i];
17.        }
18.        return maxSum;
19.    }
20.    public static void main(String[] args) {
21.        int[] arr = {1, 2, 3, 4, 5, 4, 3, 2, 1};
22.        int maxSum = getMaxEquilibriumSumOptimized(arr);
23.        System.out.println("Max Equilibrium Sum : " + maxSum);
24.    }
25. }
```



THANK YOU